CARD A01 – STRUCTURE → NAME

LEVEL : FOUNDATION **MOLECULAR FORMULA : C_6H_{14}**

1. NUMBERED PARENT



2. NAMING TRAIL

1. PARENT CHAIN: THE LONGEST CONTINUOUS CARBON CHAIN CONTAINS FIVE CARBON ATOMS; THEREFORE, THE PARENT NAME IS PENTANE.

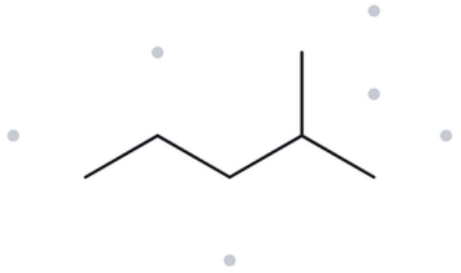
2. NUMBERING: NUMBERING FROM THE RIGHT GIVES THE METHYL SUBSTITUENT A LOCANT OF 2, WHILE NUMBERING FROM THE LEFT GIVES IT A LOCANT OF 4. THE LOWER LOCANT (2) IS SELECTED.

3. SUBSTITUENT: ONE METHYL GROUP IS ATTACHED TO CARBON 2. THUS, THE SUBSTITUENT IS IDENTIFIED AS 2-METHYL.

3. MY NAME

2-methylpentane

4. THE ALKANE CHECK



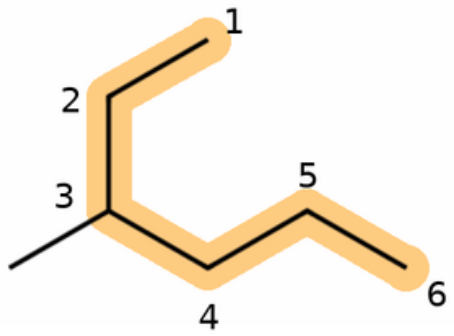
2-methylpentane **MATCH** ✓

LONGEST CHAIN CONFIRMED ✓ LOWEST LOCANTS ✓ ALPHABETICAL ORDER ✓ PUNCTUATION ✓ FORMULA COUNTED ✓

CARD A02 – STRUCTURE → NAME

LEVEL : FOUNDATION **MOLECULAR FORMULA : C_7H_{16}**

1. NUMBERED PARENT



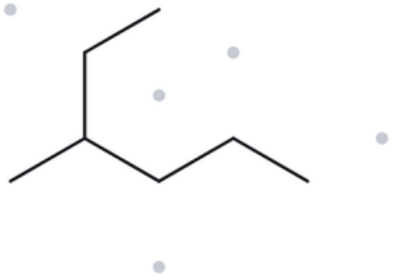
2. NAMING TRAIL

1. PARENT CHAIN: THE LONGEST CONTINUOUS CHAIN CONTAINS SIX CARBON ATOMS, SO THE PARENT NAME IS HEXANE.
2. NUMBERING: NUMBERING FROM THE UPPER END GIVES THE METHYL GROUP LOCANT 3, WHICH IS LOWER THAN 4 WHEN NUMBERED FROM THE OPPOSITE END.
3. SUBSTITUENT: A METHYL GROUP IS ATTACHED TO CARBON 3, GIVING THE PREFIX 3-METHYL.

3. MY NAME

3-methylhexane

4. THE ALKANE CHECK



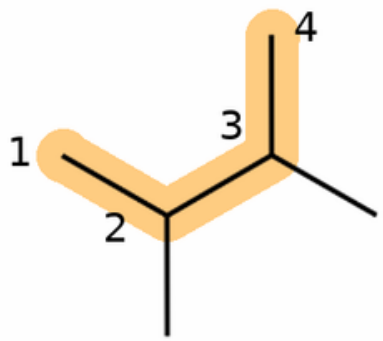
3-methylhexane **MATCH** ✓

LONGEST CHAIN CONFIRMED ✓ LOWEST LOCANTS ✓ ALPHABETICAL ORDER ✓ PUNCTUATION ✓ FORMULA COUNTED ✓

CARD A03 – STRUCTURE → NAME

LEVEL : FOUNDATION **MOLECULAR FORMULA : C_6H_{14}**

1. NUMBERED PARENT



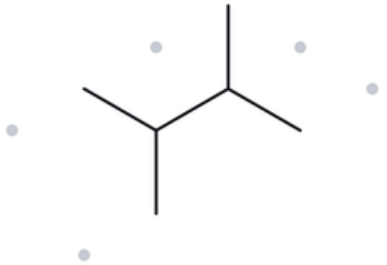
2. NAMING TRAIL

1. PARENT CHAIN: THE LONGEST CONTINUOUS CHAIN CONTAINS FOUR CARBON ATOMS, SO THE PARENT NAME IS BUTANE.
2. NUMBERING: BOTH DIRECTIONS GIVE THE SAME LOCANTS (2,3), SO EITHER DIRECTION IS ACCEPTABLE.
3. SUBSTITUENTS: TWO METHYL GROUPS ARE ATTACHED TO CARBONS 2 AND 3, GIVING THE PREFIX 2,3-DIMETHYL.

3. MY NAME

2,3-dimethylbutane

4. THE ALKANE CHECK



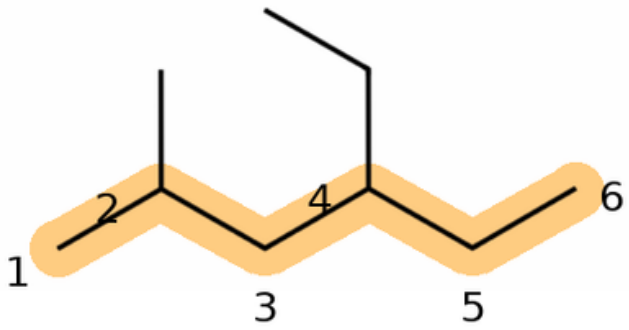
2,3-dimethylbutane **MATCH** ✓

LONGEST CHAIN CONFIRMED ✓ LOWEST LOCANTS ✓ ALPHABETICAL ORDER ✓ PUNCTUATION ✓ FORMULA COUNTED ✓

CARD A04 – STRUCTURE → NAME

LEVEL : DEVELOPING **MOLECULAR FORMULA : C_9H_{20}**

1. NUMBERED PARENT



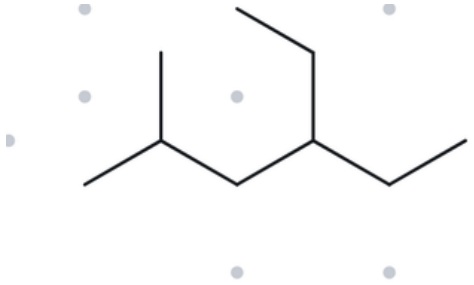
2. NAMING TRAIL

1. PARENT CHAIN: THE LONGEST CONTINUOUS CHAIN CONTAINS SIX CARBON ATOMS, SO THE PARENT NAME IS HEXANE.
2. NUMBERING: NUMBERING FROM THE LEFT GIVES LOCANTS 2 AND 4, WHICH ARE LOWER THAN 3 AND 5 FROM THE OPPOSITE DIRECTION.
3. SUBSTITUENTS: AN ETHYL GROUP IS ATTACHED TO C-4 AND A METHYL GROUP TO C-2. THEY ARE ARRANGED ALPHABETICALLY, WITH ETHYL BEFORE METHYL.

3. MY NAME

4-ethyl-2-methylhexane

4. THE ALKANE CHECK



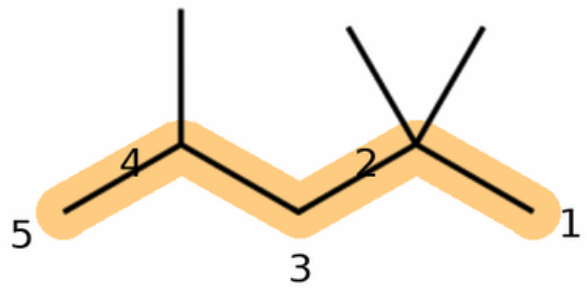
4-ethyl-2-methylhexane **MATCH** ✓

LONGEST CHAIN CONFIRMED ✓ LOWEST LOCANTS ✓ ALPHABETICAL ORDER ✓ PUNCTUATION ✓ FORMULA COUNTED ✓

CARD A05 – STRUCTURE → NAME

LEVEL : DEVELOPING **MOLECULAR FORMULA : C_8H_{18}**

1. NUMBERED PARENT



2. NAMING TRAIL

1. PARENT CHAIN: THE LONGEST CONTINUOUS CHAIN CONTAINS FIVE CARBON ATOMS, SO THE PARENT NAME IS PENTANE.

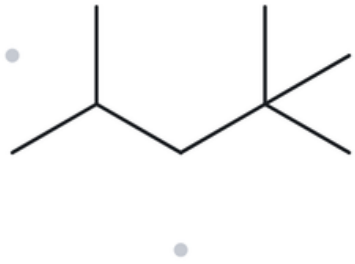
2. NUMBERING: NUMBERING FROM THE RIGHT GIVES LOCANTS 2,2,4, WHICH ARE LOWER THAN 2,4,4 FROM THE OPPOSITE DIRECTION.

3. SUBSTITUENTS: THREE METHYL GROUPS ARE ATTACHED TO C-2, C-2, AND C-4, GIVING THE PREFIX 2,2,4-TRIMETHYL.

3. MY NAME

2,2,4-trimethylpentane

4. THE ALKANE CHECK



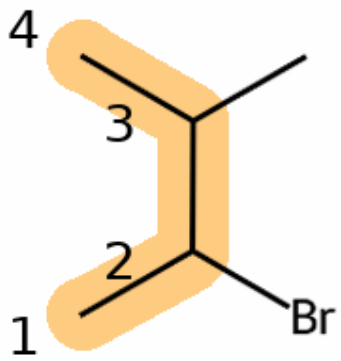
2,2,4-trimethylpentane **MATCH** ✓

LONGEST CHAIN CONFIRMED ✓ LOWEST LOCANTS ✓ ALPHABETICAL ORDER ✓ PUNCTUATION ✓ FORMULA COUNTED ✓

CARD A06 – STRUCTURE → NAME

LEVEL : DEVELOPING **MOLECULAR FORMULA : $C_5H_{11}BR$**

1. NUMBERED PARENT



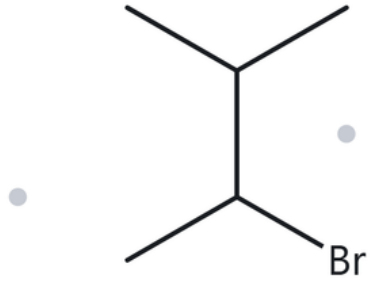
2. NAMING TRAIL

1. PARENT CHAIN: THE LONGEST CONTINUOUS CHAIN CONTAINS FOUR CARBON ATOMS, SO THE PARENT NAME IS BUTANE.
2. NUMBERING: BOTH DIRECTIONS GIVE LOCANTS 2 AND 3. SINCE THE LOCANTS ARE TIED, THE BROMO SUBSTITUENT GETS THE LOWER NUMBER BECAUSE IT COMES FIRST ALPHABETICALLY.
3. SUBSTITUENTS: A BROMO GROUP IS ATTACHED TO C-2 AND A METHYL GROUP TO C-3.

3. MY NAME

2-bromo-3-methylbutane

4. THE ALKANE CHECK



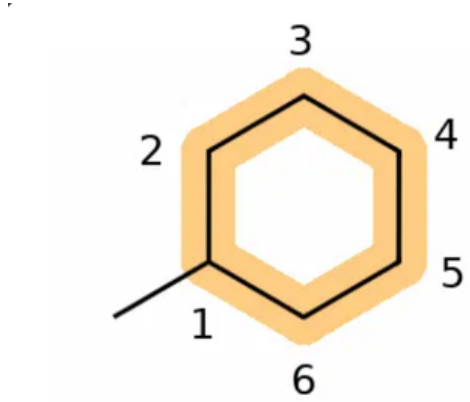
2-bromo-3-methylbutane **MATCH** ✓

LONGEST CHAIN CONFIRMED ✓ LOWEST LOCANTS ✓ ALPHABETICAL ORDER ✓ PUNCTUATION ✓ FORMULA COUNTED ✓

CARD A07 – STRUCTURE → NAME

LEVEL : DEVELOPING **MOLECULAR FORMULA : C_7H_{14}**

1. NUMBERED PARENT



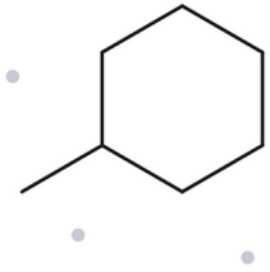
2. NAMING TRAIL

1. PARENT CHAIN: THE STRUCTURE CONTAINS A SIX-CARBON RING, SO THE PARENT NAME IS CYCLOHEXANE.
2. NUMBERING: THE CARBON BEARING THE METHYL GROUP IS ASSIGNED LOCANT 1. NO DIRECTION COMPARISON IS NEEDED SINCE THERE IS ONLY ONE SUBSTITUENT.
3. SUBSTITUENT: ONE METHYL GROUP IS ATTACHED TO THE RING, GIVING THE NAME METHYLCYCLOHEXANE.

3. MY NAME

methylcyclohexane

4. THE ALKANE CHECK



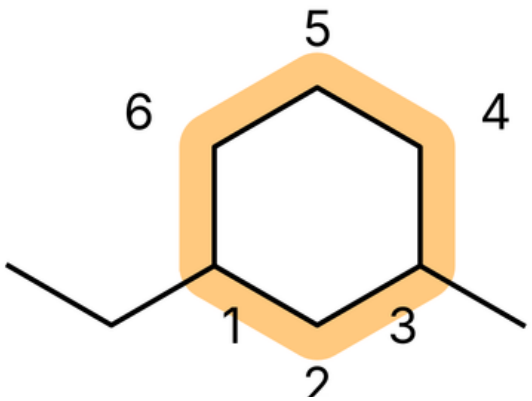
methylcyclohexane **MATCH** ✓

LONGEST CHAIN CONFIRMED ✓ LOWEST LOCANTS ✓ ALPHABETICAL ORDER ✓ PUNCTUATION ✓ FORMULA COUNTED ✓

CARD A08 – STRUCTURE → NAME

LEVEL : CHALLENGE **MOLECULAR FORMULA : C_9H_{18}**

1. NUMBERED PARENT



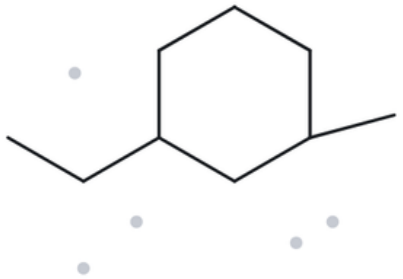
2. NAMING TRAIL

1. PARENT CHAIN: THE STRUCTURE CONTAINS A SIX-CARBON RING, SO THE PARENT NAME IS CYCLOHEXANE.
2. NUMBERING: BOTH DIRECTIONS GIVE LOCANTS 1 AND 3. ETHYL GETS LOCANT 1 BECAUSE IT COMES FIRST ALPHABETICALLY.
3. SUBSTITUENTS: AN ETHYL GROUP IS ATTACHED TO C-1 AND A METHYL GROUP TO C-3.

3. MY NAME

1-ethyl-3-methylcyclohexane

4. THE ALKANE CHECK



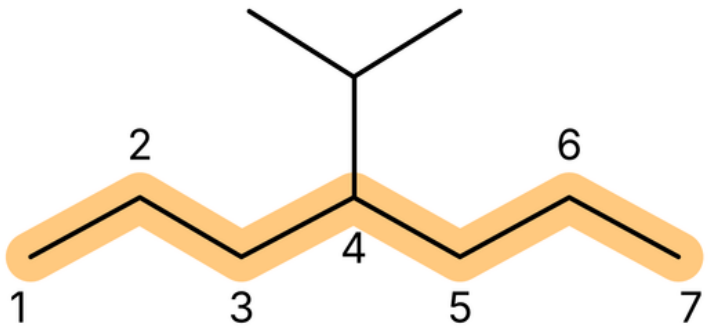
1-ethyl-3-methylcyclohexane **MATCH** ✓

LONGEST CHAIN CONFIRMED ✓ LOWEST LOCANTS ✓ ALPHABETICAL ORDER ✓ PUNCTUATION ✓ FORMULA COUNTED ✓

CARD A09 – STRUCTURE → NAME

LEVEL : CHALLENGE **MOLECULAR FORMULA : $C_{10}H_{22}$**

1. NUMBERED PARENT



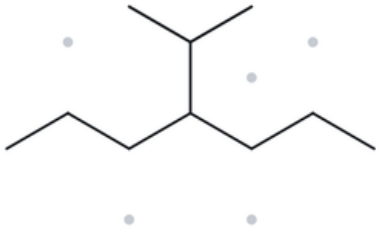
2. NAMING TRAIL

1. PARENT CHAIN: THE LONGEST CONTINUOUS CHAIN CONTAINS SEVEN CARBON ATOMS, SO THE PARENT NAME IS HEPTANE.
2. NUMBERING: BOTH DIRECTIONS GIVE THE SAME LOCANT, 4, FOR THE SUBSTITUENT.
3. SUBSTITUENT: AN ISOPROPYL GROUP IS ATTACHED TO C–4, GIVING THE NAME 4–ISOPROPYL.

3. MY NAME

4-isopropylheptane

4. THE ALKANE CHECK



2-methyl-3-propylhexane **MISMATCHED ✗**

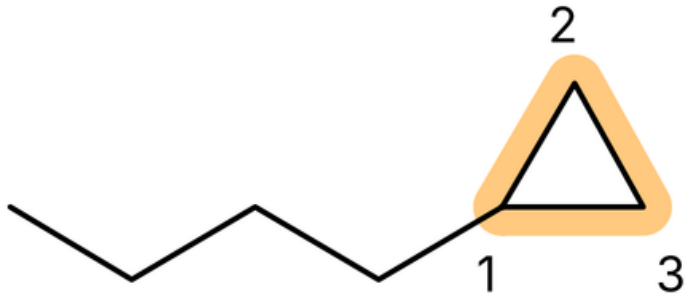
SAME STRUCTURE, BUT NOT THE LONGEST CHAIN (6 VS 7 C)

LONGEST CHAIN CONFIRMED ✓ LOWEST LOCANTS ✓ ALPHABETICAL ORDER ✓ PUNCTUATION ✓ FORMULA COUNTED ✓

CARD A10 – STRUCTURE → NAME

LEVEL : CHALLENGE **MOLECULAR FORMULA : C_7H_{14}**

1. NUMBERED PARENT



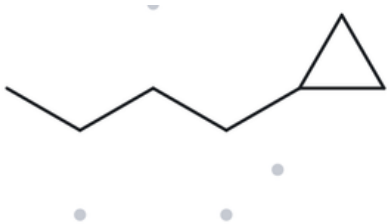
2. NAMING TRAIL

1. PARENT RING: THE STRUCTURE CONTAINS A THREE-CARBON RING, SO THE PARENT NAME IS CYCLOPROPANE.
2. NUMBERING: THE RING HAS ONLY ONE SUBSTITUENT, SO THE ATTACHMENT POINT IS AUTOMATICALLY ASSIGNED LOCANT 1.
3. SUBSTITUENT: A FOUR-CARBON CHAIN IS ATTACHED TO THE RING, GIVING THE SUBSTITUENT NAME BUTYL.

3. MY NAME

butylcyclopropane

4. THE ALKANE CHECK



butylcyclopropane **MATCH** ✓

LONGEST CHAIN CONFIRMED ✓ LOWEST LOCANTS ✓ ALPHABETICAL ORDER ✓ PUNCTUATION ✓ FORMULA COUNTED ✓

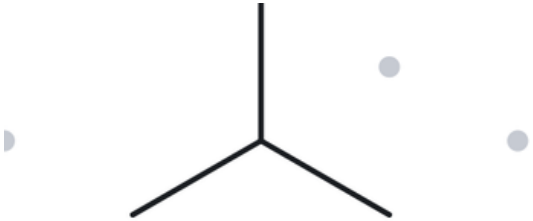
CARD BOT – NAME → STRUCTURE

LEVEL : FOUNDATION **GIVEN NAME : 2-METHYLPROPANE**

1. NAME DECODED

- PARENT: PROPANE → A 3-CARBON CHAIN.
- SUBSTITUENT: METHYL GROUP ATTACHED TO C-2.
- EXPECTED FORMULA: 3 + 1 = 4 CARBONS; THEREFORE, C_4H_{10}.

2.BOND-LINE STRUCTURE (THE ALKANE)



2-methylpropane

3. FORMULA CHECK

- COUNTED: 4 CARBON ATOMS AND 10 HYDROGEN ATOMS.
- ALKANE FORMULA: C_nH_{2n+2} = C_4H_{10} ✓

4.BACK-CHECK

- EXPECTED RETURNED NAME: 2-METHYLPROPANE.
- SAME AS THE GIVEN NAME ✓

PARENT LENGTH MATCHES ✓ SUBSTITUENT POSITIONS MATCH ✓ FORMULA AGREES ✓ BACK-CHECK RECORDED ✓

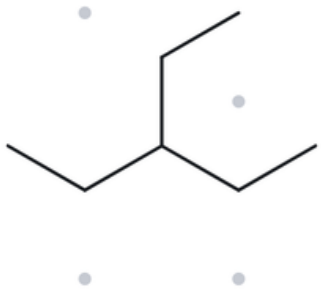
CARD B02 – NAME → STRUCTURE

LEVEL : FOUNDATION GIVEN NAME : 3-ETHYLPENTANE

1. NAME DECODED

- PARENT: PENTANE → A 5-CARBON CHAIN.
- SUBSTITUENT: ONE ETHYL GROUP ATTACHED TO C-3.
- EXPECTED FORMULA: 5 + 2 = 7 CARBONS; THEREFORE, C_7H_{16}.

2.BOND-LINE STRUCTURE (THE ALKANE)



3-ethylpentane

3. FORMULA CHECK

- COUNTED FROM THE NAME:
  C_7H_{16} – 7 CARBON ATOMS AND 16 HYDROGEN ATOMS.
- ALKANE:
  C_nH_{2n+2} = C7H16 ✓

4.BACK-CHECK

- EXPECTED RETURNED NAME: 3-ETHYLPENTANE.
- SAME AS THE GIVEN NAME ✓

PARENT LENGTH MATCHES ✓ SUBSTITUENT POSITIONS MATCH ✓ FORMULA AGREES ✓ BACK-CHECK RECORDED ✓

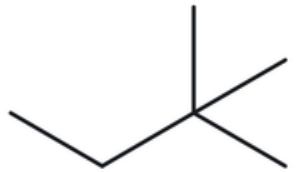
CARD B03 – NAME → STRUCTURE

LEVEL : FOUNDATION **GIVEN NAME : 2,2-DIMETHYLBUTANE**

1. NAME DECODED

- PARENT: BUTANE → A 4-CARBON CHAIN.
- SUBSTITUENTS: TWO METHYL GROUPS ATTACHED TO C-2.
- EXPECTED FORMULA: 4 + 2 = 6 CARBONS; THEREFORE, C_6H_{14}.

2.BOND-LINE STRUCTURE (THE ALKANE)



2,2-dimethylbutane

3. FORMULA CHECK

- COUNTED FROM THE NAME: C_6H_{14} – 6 CARBON ATOMS AND 14 HYDROGEN ATOMS.
- ALKANE:
  C_nH_{2n+2} = C_6H_{14} ✓

4.BACK-CHECK

- EXPECTED RETURNED NAME: 2,2-DIMETHYLBUTANE
- ✓ SAME AS THE GIVEN NAME.

PARENT LENGTH MATCHES ✓ SUBSTITUENT POSITIONS MATCH ✓ FORMULA AGREES ✓ BACK-CHECK RECORDED ✓

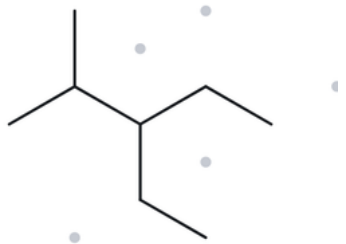
CARD B04 – NAME → STRUCTURE

LEVEL : DEVELOPING **GIVEN NAME : 3-ETHYL-2-METHYLPENTANE**

1. NAME DECODED

- PARENT: PENTANE → A 5-CARBON CHAIN.
- SUBSTITUENTS: ETHYL GROUP ON C-3 AND METHYL GROUP ON C-2.
- EXPECTED FORMULA: 5 + 2 + 1 = 8 CARBONS; THEREFORE, C_8H_{18}.

2.BOND-LINE STRUCTURE (THE ALKANE)



3-ethyl-2-methylpentane

3. FORMULA CHECK

- COUNTED FROM THE NAME: C_8H_{18} – 8 CARBON ATOMS AND 18 HYDROGEN ATOMS.
- ALKANE:
- C_nH_{2n+2} = C8H18 ✓

4.BACK-CHECK

- EXPECTED RETURNED NAME: 3-ETHYL-2-METHYLPENTANE
- ✓ SAME AS THE GIVEN NAME.

PARENT LENGTH MATCHES ✓ SUBSTITUENT POSITIONS MATCH ✓ FORMULA AGREES ✓ BACK-CHECK RECORDED ✓

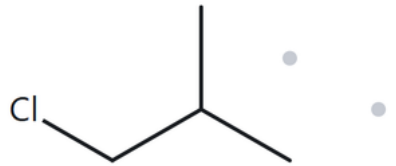
CARD B05 – NAME → STRUCTURE

LEVEL : DEVELOPING **GIVEN NAME : 1-CHLORO-2-METHYLPROPANE**

1. NAME DECODED

- PARENT: PROPANE → A 3-CARBON CHAIN.
- SUBSTITUENTS: ONE CHLORO GROUP ON C-1 AND ONE METHYL GROUP ON C-2.
- EXPECTED FORMULA: 3 + 1 = 4 CARBONS; THEREFORE, C_4H_9CL.

2.BOND-LINE STRUCTURE (THE ALKANE)



1-chloro-2-methylpropane

3. FORMULA CHECK

- COUNTED FROM THE NAME: C_4H_9CL → 4 CARBON ATOMS, 9 HYDROGEN ATOMS, AND 1 CHLORINE ATOM.
- PARENT ALKANE: C_nH_{2n+2} = C_4H_{10}; REPLACE 1 H WITH CL = C_4H_9CL ✓

4.BACK-CHECK

- EXPECTED RETURNED NAME: 1-CHLORO-2-METHYLPROPANE
- ✓ SAME AS THE GIVEN NAME.

PARENT LENGTH MATCHES ✓ SUBSTITUENT POSITIONS MATCH ✓ FORMULA AGREES ✓ BACK-CHECK RECORDED ✓

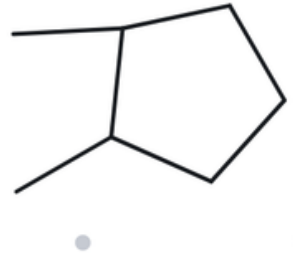
CARD B06 – NAME → STRUCTURE

LEVEL : DEVELOPING GIVEN NAME : 1,2-DIMETHYLCYCLOPENTANE

1. NAME DECODED

- PARENT: CYCLOPENTANE → A 5-CARBON RING.
- SUBSTITUENTS: TWO METHYL GROUPS, ONE ON C-1 AND ONE ON C-2.
- EXPECTED FORMULA: 5 + 2 = 7 CARBONS; THEREFORE, C_7H_{14}.

2.BOND-LINE STRUCTURE (THE ALKANE)



1,2-dimethylcyclopentane

3. FORMULA CHECK

- COUNTED FROM THE NAME: C_7H_{14} → 7 CARBON ATOMS AND 14 HYDROGEN ATOMS.
- CYCLOALKANE: C_nH_{2n} = C_7H_{14} ✓

4.BACK-CHECK

- EXPECTED RETURNED NAME: 1,2-DIMETHYLCYCLOPENTANE
- ✓ SAME AS THE GIVEN NAME.

PARENT LENGTH MATCHES ✓ SUBSTITUENT POSITIONS MATCH ✓ FORMULA AGREES ✓ BACK-CHECK RECORDED ✓

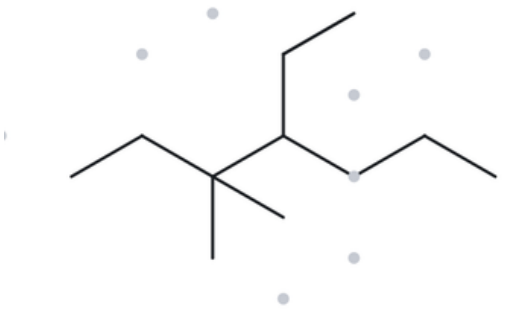
CARD B07 – NAME → STRUCTURE

LEVEL : DEVELOPING GIVEN NAME : 4-ETHYL-3,3-DIMETHYLHEPTANE

1. NAME DECODED

- PARENT: HEPTANE → A 7-CARBON CHAIN.
- SUBSTITUENTS: TWO METHYL GROUPS ON C-3 AND ONE ETHYL GROUP ON C-4.
- EXPECTED FORMULA: 7 + 2 + 2 = 11 CARBONS; THEREFORE, $C_{11}H_{24}$.

2.BOND-LINE STRUCTURE (THE ALKANE)



4-ethyl-3,3-dimethylheptane

3. FORMULA CHECK

- COUNTED FROM THE NAME: $C_{11}H_{24}$ → 11 CARBON ATOMS AND 24 HYDROGEN ATOMS.
- ALKANE: C_nH_{2n+2} = $C_{11}H_{24}$ ✓

4.BACK-CHECK

- EXPECTED RETURNED NAME: 4-ETHYL-3,3-DIMETHYLHEPTANE
- ✓ SAME AS THE GIVEN NAME.

PARENT LENGTH MATCHES ✓ SUBSTITUENT POSITIONS MATCH ✓ FORMULA AGREES ✓ BACK-CHECK RECORDED ✓

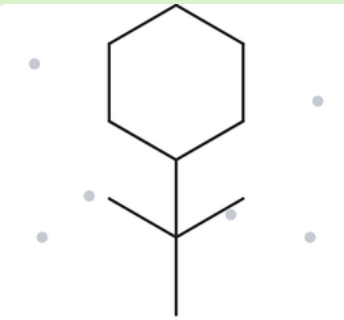
CARD B08 – NAME → STRUCTURE

LEVEL : CHALLENGE **GIVEN NAME : TERT-BUTYLCYCLOHEXANE**

1. NAME DECODED

- PARENT: CYCLOHEXANE → A 6-CARBON RING.
- SUBSTITUENTS: ONE TERT-BUTYL GROUP, $-C(CH_3)_3$, ATTACHED TO C-1.
- EXPECTED FORMULA: 6 + 4 = 10 CARBONS; THEREFORE, $C_{10}H_{20}$.

2.BOND-LINE STRUCTURE (THE ALKANE)



(1,1-dimethylethyl)cyclohexane

3. FORMULA CHECK

- COUNTED FROM THE NAME: $C_{10}H_{20}$ → 10 CARBON ATOMS AND 20 HYDROGEN ATOMS.
- CYCLOALKANE RING: C_6H_{12}; REPLACE 1 H WITH C_4H_9 = $C_{10}H_{20}$ ✓

4.BACK-CHECK

- EXPECTED RETURNED NAME: TERT-BUTYLCYCLOHEXANE
- RETURNED: (1,1-DIMETHYLETHYL)CYCLOHEXANE
- ✓ SAME AS THE GIVEN NAME.

PARENT LENGTH MATCHES ✓ SUBSTITUENT POSITIONS MATCH ✓ FORMULA AGREES ✓ BACK-CHECK RECORDED ✓

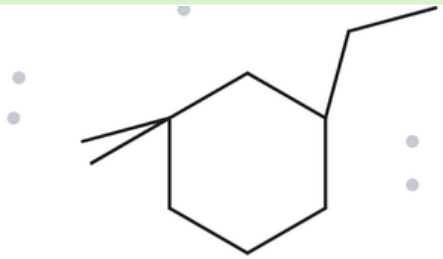
CARD B09 – NAME → STRUCTURE

LEVEL : CHALLENGE **GIVEN NAME : 3-ETHYL-1,1 DIMETHYLCYCLOHEXANE**

1. NAME DECODED

- PARENT: CYCLOHEXANE → A 6-CARBON RING.
- SUBSTITUENTS: TWO METHYL GROUPS ON C-1 AND ONE ETHYL GROUP ON C-3.
- EXPECTED FORMULA: 6 + 2 + 2 = 10 CARBONS; THEREFORE, $C_{10}H_{20}$.

2.BOND-LINE STRUCTURE (THE ALKANE)



3-ethyl-1,1-dimethylcyclohexane

3. FORMULA CHECK

- COUNTED FROM THE NAME: $C_{10}H_{20}$ → 10 CARBON ATOMS AND 20 HYDROGEN ATOMS.
- CYCLOALKANE RING: C_6H_{12}; REPLACE 3 H WITH 2 CH_3 AND 1 C_2H_5 = $C_{10}H_{20}$ ✓

4.BACK-CHECK

- EXPECTED RETURNED NAME: 3-ETHYL-1,1-DIMETHYLCYCLOHEXANE
- ✓ SAME AS THE GIVEN NAME.

PARENT LENGTH MATCHES ✓ SUBSTITUENT POSITIONS MATCH ✓ FORMULA AGREES ✓ BACK-CHECK RECORDED ✓

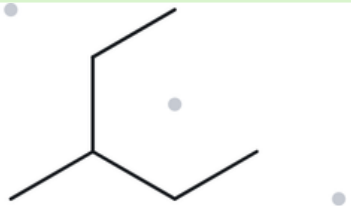
CARD B10 – NAME → STRUCTURE

LEVEL : CHALLENGE GIVEN NAME : 2-ETHYLBUTANE

1. NAME DECODED

- PARENT: BUTANE → A 4-CARBON CHAIN.
- SUBSTITUENTS: ONE ETHYL GROUP ATTACHED TO C-2.
- EXPECTED FORMULA: 4 + 2 = 6 CARBONS; THEREFORE, C_6H_{14}.

2.BOND-LINE STRUCTURE (THE ALKANE)



3-methylpentane

3. FORMULA CHECK

- COUNTED FROM THE NAME: C_6H_{14} → 6 CARBON ATOMS AND 14 HYDROGEN ATOMS.
- ALKANE: C_nH_{2n+2} = C_6H_{14} ✓

4.BACK-CHECK

- EXPECTED RETURNED NAME: 3-METHYLPENTANE
- ✗ NOT THE SAME AS THE GIVEN NAME. THE LONGEST CHAIN IS 5 CARBONS (PENTANE), SO THE ETHYL GROUP BECOMES PART OF THE MAIN CHAIN.

PARENT LENGTH MATCHES ✗ SUBSTITUENT POSITIONS MATCH ✗ FORMULA AGREES ✓ BACK-CHECK RECORDED ✓